

Take off 10:32 UAK → QAS-U ^{note} = 2 visits

Maintenance sheet version 10 for ablation area (tripod, one level)

M. Tranter
made new
PROMBIO
measurement
with chissel
and random
spatial
sampling and
took cores

Station name	QAS_U_v3
People present	JEB, JJ, pilot siggi serneg
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	27 Aug 2024 + 1 sept.

Notes:

install snow Fox upgrade; wind
T string not working
fair weather, return trip fair wx

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	/		213257	Clean? (Y) N
Ablation hose assembly	N	/		5017	No air in hose and bladder half full? Y/N
Iridium modem IMEI number	N		21950	21950	
Inclinometer/compass	N	N	Attached to the radiometer? (Y) N inside		
Satellite antenna	N	N			
Wind sensor	N	N	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N	N	Fan spins? (Y) N Sounds, OK? (Y) N		
HygroClip T/hum assembly	N	N	NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	(Y)	✓	Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	(Y)	✓	Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string And maybe an ablobender	N	/			
Solar panel	N	/	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N	/	At top of logger enclosure? Y / N		
Logger box ID	N	/	CR1000 or CR1000X? ID number:		
Multiplexer	N	/	Clicking during measurement cycle? Y / N		
Logger enclosure	N	/	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N	/	New box? Y / N New desiccant bags? Y / N		
Stakes	(Y)	/	New one(s) drilled in? (Y) N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N	/	New tripod? (Y) N New sensor boom? Y / N		

existing is in
3.59 cm

Station name	QAS_U_v3
Date	27 Aug 2024

050502 → latest

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+45 sec	
Download data and/or change CF/SD Card (wait for green light)	Y/N	Y/N
Name of logger program	Program revised	1.5
Screen dump of values in fast scan mode	V3 1.4 Y/N	☒ Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	33 ☒ Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	4 °	4 °
Mast tilt across boom looking from radiometer (- if clockwise)	+5 °	-5 °
Boom direction relative to north	178 ° true/magn.	Same ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	☒ Y/N	☒ Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	☒ Y/N	☒ Y/N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	440 cm	440 cm
Free length of stake 2 (if 3: middle)	! see cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	below * cm	cm
Height of ablation bladder fluid level above the ice surface	173 cm	173 cm
Vertical difference in surface height (at station – at hose) (approx.)	-5 cm	-5 cm
Length of thermistor string on surface from surface marking	±52.5 cm	same cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

- stake assem not re drilled
- * 2 free old stakes, 1 to W2.66.7 cm top of carbon
- # 2.286.9 to top
- ± 1 thermistor is out
- Must-do at next visit: photos document 2 x PTA (old)
- ~~cross mount replacement for SNOW FOX. we had to improvise that piece~~
- replace at least 1 leg (bent)
- there are pipes laying around @ & @ QAS-B

We made it right 1 Sept
see Snow Fox
Maintenance Sheet